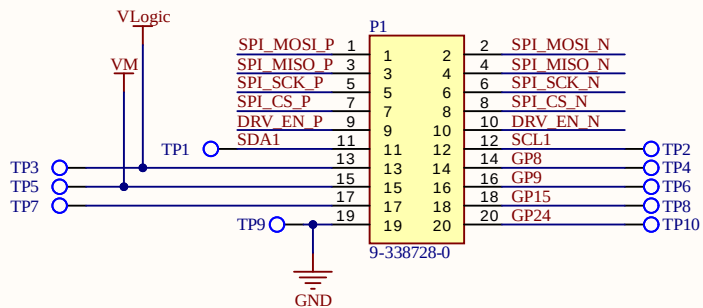


Baseboard Interface

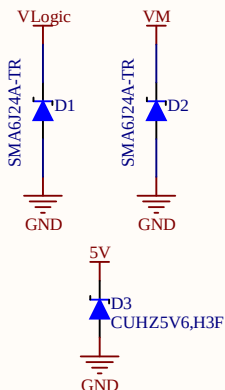
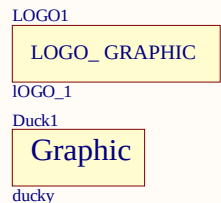


Release Notes

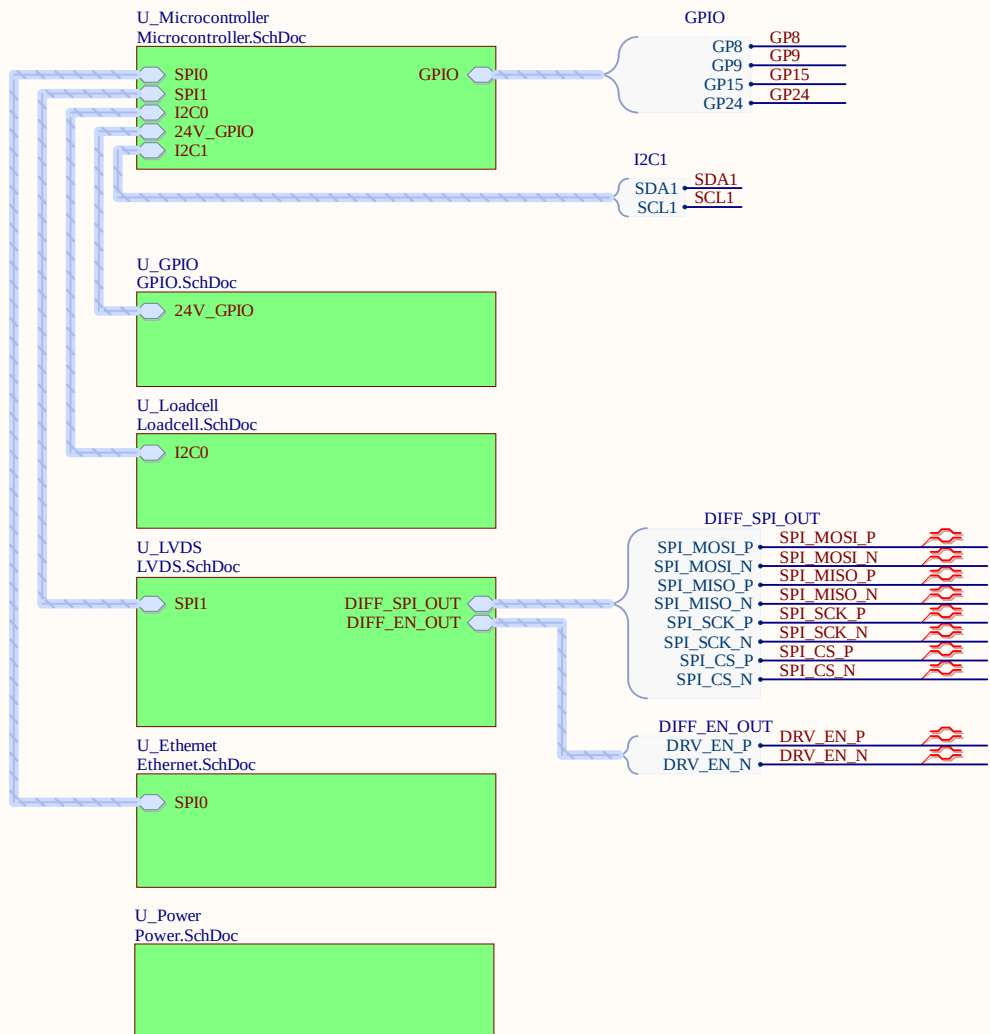
R001: Initial Release

R002:
Fix LED placement
add configurable ADIN SPI modesR003:
Remove Switching RegulatorsR004:
Replace Switching Regulators
Add debug port
Add Differential I2CR005:
Fix Differential I2C Buffer Power Net

Misc



Top Level



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SHEET 1 OF 7

R5

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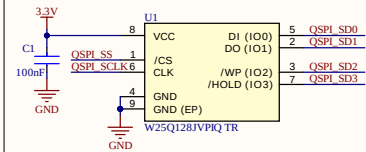
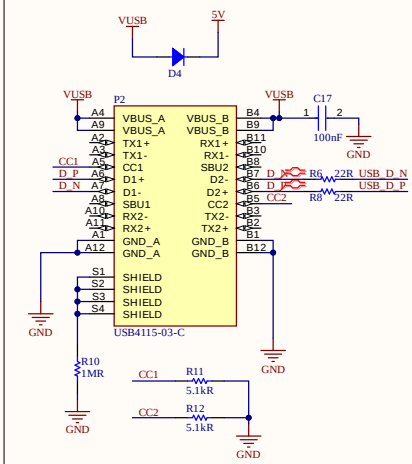
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Program flash

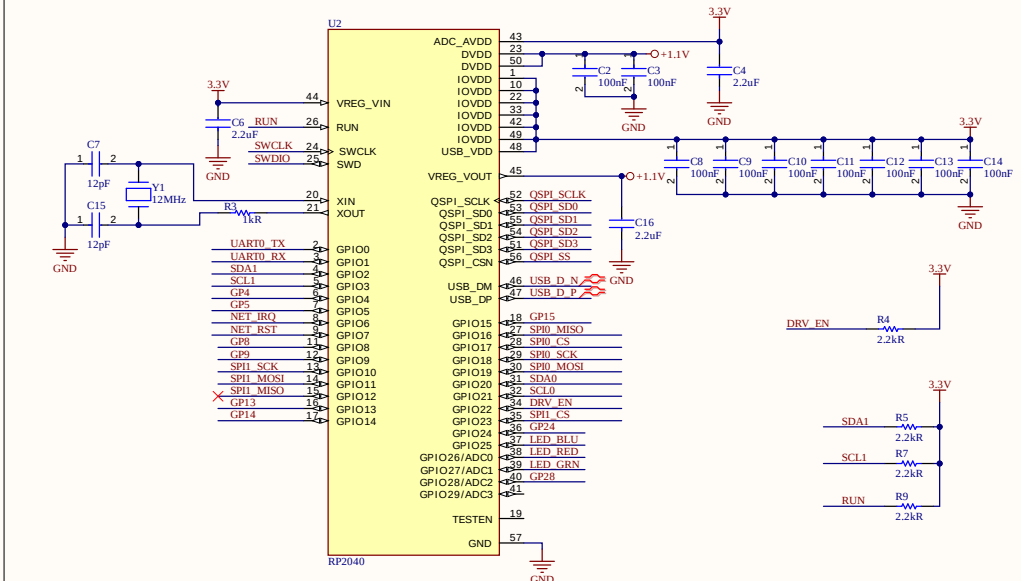
The diagram shows the connection of a W25Q128JVPIQTR flash memory (U1) for programming. The chip is connected to a 3.3V supply and ground. The connections are as follows:

- Pin 8 (VCC) is connected to 3.3V.
- Pin 9 (GND) is connected to GND.
- Pin 1 (QSPI_SS) is connected to 3.3V through capacitor C1.
- Pin 6 (QSPI_SCLK) is connected to 3.3V.
- Pin 5 (QSPI_SD0) is connected to pin 2 (QSPI_SD1).
- Pin 3 (QSPI_SD2) is connected to pin 7 (QSPI_SD3).

The chip is labeled U1 and W25Q128JVPIQTR.

[illegible]

The schematic diagram illustrates the RP2040 Microcontroller board. The central component is the RP2040 chip (U2), which is connected to various external components. The power supply section includes a 3.3V input (VREG_VIN) connected to the SWCLK, SWDIO, and RUN pins. A 1.1V regulator output (VREG_VOUT) is connected to the QSPI_SCLK and QSPI_SD0 pins. The USB interface is shown with D+ and D- pins connected to a USB connector. The UART interface includes TX and RX pins connected to a 12MHz crystal and capacitors. The SPI interface includes SCK, MOSI, and MISO pins connected to a 12pF capacitor. Other pins are connected to ground or specific components like resistors and capacitors. The diagram is labeled with component values and pin numbers.

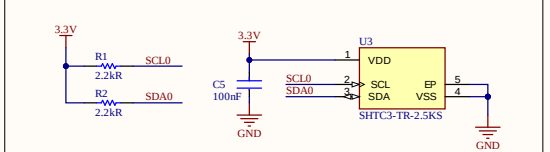


Temp/Humidity Sensor

The diagram illustrates the wiring for a Temp/Humidity Sensor (U3, SHTC3-TR-2.5KS) connected to a 3.3V supply and GND. The sensor's pins are connected as follows:

- Pin 1 (VDD):** Connected to the 3.3V supply.
- Pin 2 (SCL):** Connected to the SCL0 line, which is pulled up to 3.3V by resistor R1 (2.2kΩ).
- Pin 3 (SDA):** Connected to the SDA0 line, which is pulled up to 3.3V by resistor R2 (2.2kΩ).
- Pin 4 (EP):** Connected to GND.
- Pin 5 (VSS):** Connected to GND.

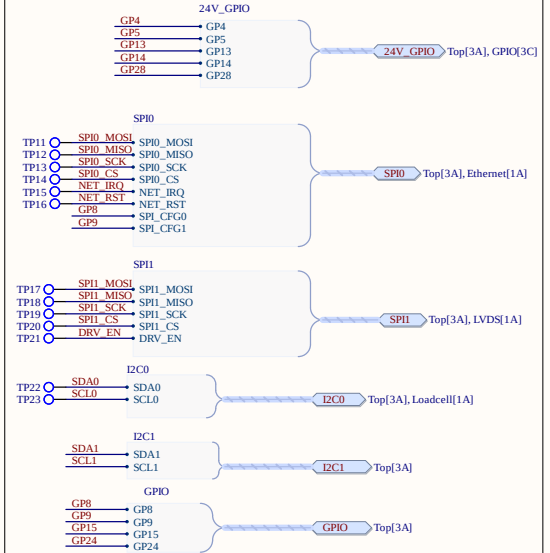
A 100nF capacitor (C5) is connected between the 3.3V supply and GND for decoupling.



Harnesses

The diagram illustrates various harnesses connected to a central system. The harnesses are categorized by protocol:

- 24V_GPIO:** GP4, GP5, GP13, GP14, GP28. Connected to Top[3A], GPIO[3C].
- SPI0:** SPI0_MOSI, SPI0_MISO, SPI0_SCK, SPI0_CS, NET_IRQ, NET_RST, GP8, GP9. Connected to Top[3A], Ethernet[1A].
- SPI1:** SPI1_MOSI, SPI1_MISO, SPI1_SCK, SPI1_CS, DRV_EN. Connected to Top[3A], LVDS[1A].
- I2C0:** SDA0, SCL0. Connected to Top[3A], LoadCell[1A].
- I2C1:** SDA1, SCL1. Connected to Top[3A].
- GPIO:** GP6, GP9, GP15, GP24. Connected to Top[3A].



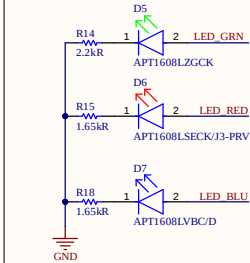
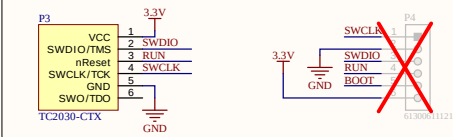
Buttons

The image shows two circuit diagrams for buttons, labeled BOOTSEL and RESET. Both buttons are SPST (Single Pole Single Throw) switches with pins 1, 2, 3, and 4. Pin 1 is connected to GND. Pin 2 is connected to the other side of the switch. Pin 3 is connected to a pull-up resistor (R13, 1kR) which is connected to the VCC line (labeled BOOT for BOOTSEL and RESET for RESET). Pin 4 is connected to the other side of the switch. The output of the switch is connected to the QSPI_SS pin for BOOTSEL and the RUN pin for RESET.



Programming/Debug

The image displays two circuit diagrams for programming and debugging. The left diagram shows the correct connections for the TC2030-CTX: VCC to 3.3V, SWDIO to pin 2, RUN to pin 3, SWCLK to pin 4, SWCLK/ICLK to pin 5, and SWO/TDO to pin 6, all connected to GND. The right diagram shows the connections for the P4 (6185001121) which are crossed out with a large red X, indicating they are not applicable for the TC2030-CTX. The P4 connections would be SWCLK to pin 3, SWDIO to pin 4, RUN to pin 5, and BOOT to pin 6, with a 3.3V supply to pin 1 and GND to pin 2.

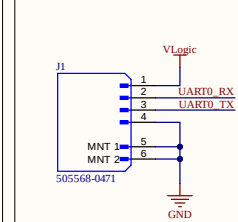
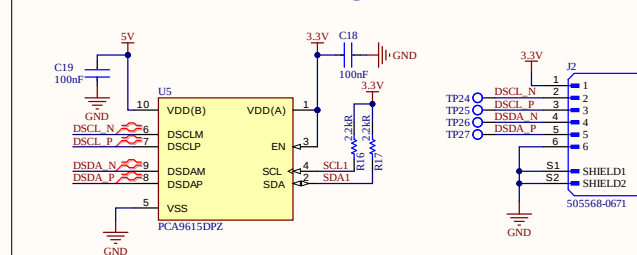


Serial/Sensor

Wiring diagram for the Serial/Sensor module:

- Pin 1: VLogic
- Pin 2: UART0_RX
- Pin 3: UART0_TX
- Pin 4: GND
- Pin 5: MNT
- Pin 6: GND

Connector: 505568-0471

[illegible]

The diagram shows the pin configuration for the NVRAM chip U4 (MB85RC64TAPN-G-AMEWE1). The chip is represented as a rectangle with pins numbered 1 through 9. Pin 1 is connected to VDD. Pin 2 is connected to A0. Pin 3 is connected to A1. Pin 4 is connected to A2. Pin 5 is connected to SCL. Pin 6 is connected to SDA. Pin 7 is connected to WP. Pin 8 is connected to EP. Pin 9 is connected to VSS. A 3.3V supply is connected to pin 8 through a 100nF capacitor. The capacitor is labeled C20. The supply is also connected to pin 4. The ground connections are labeled GND.

U4 MB85RC64TAPN-G-AMEWE1

3.3V

C20

100nF

1 A0 VDD

2 A1

3 A2 SCL

4 A2 WP SDA

5 EP VSS

6

7

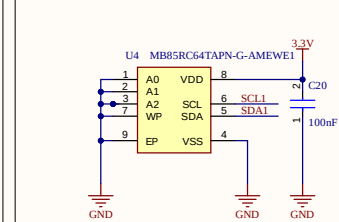
8

9

GND

GND

GND

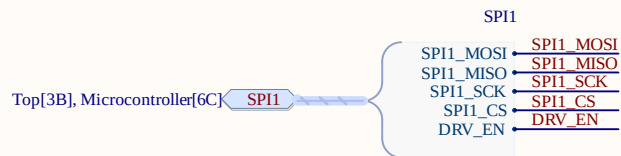


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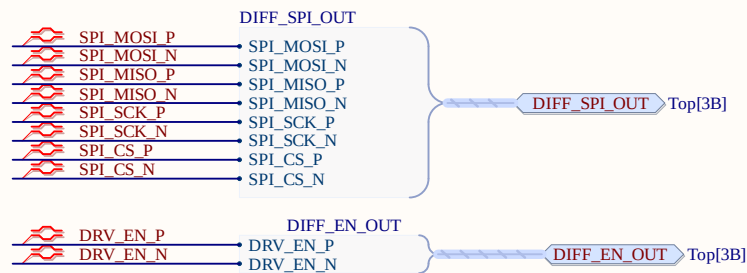
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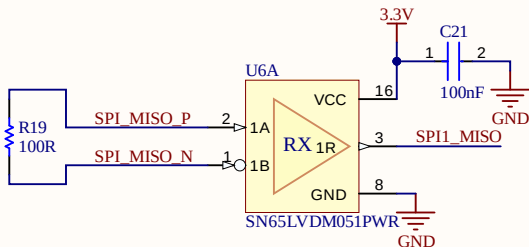
Single-Ended SPI



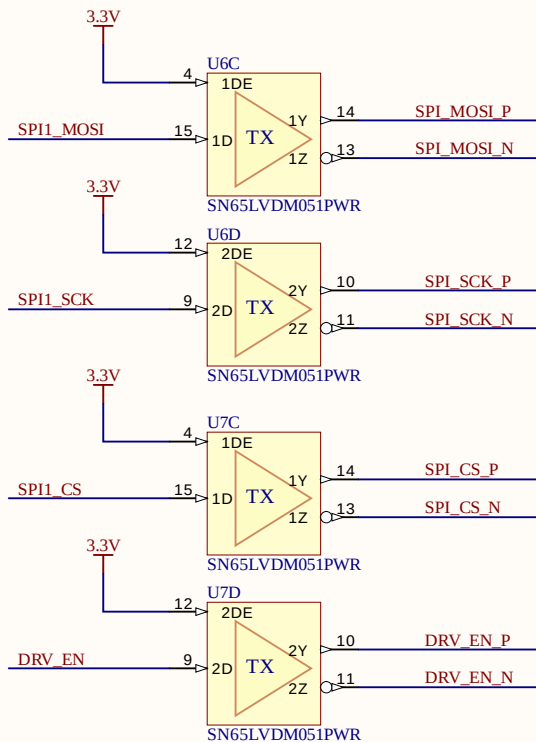
Differential SPI



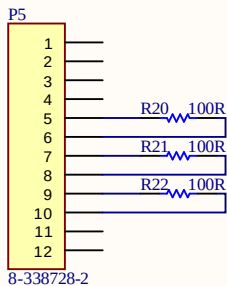
Receiver



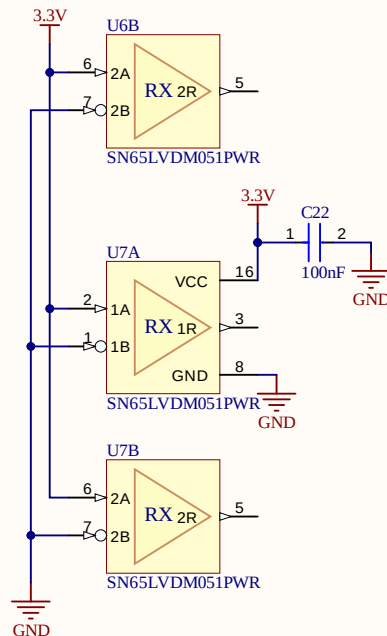
Transmitters



Terminator



Unused



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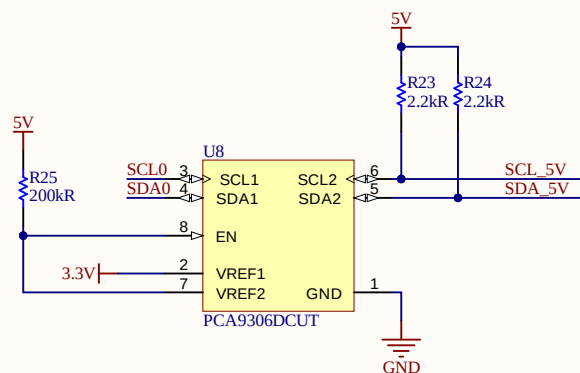
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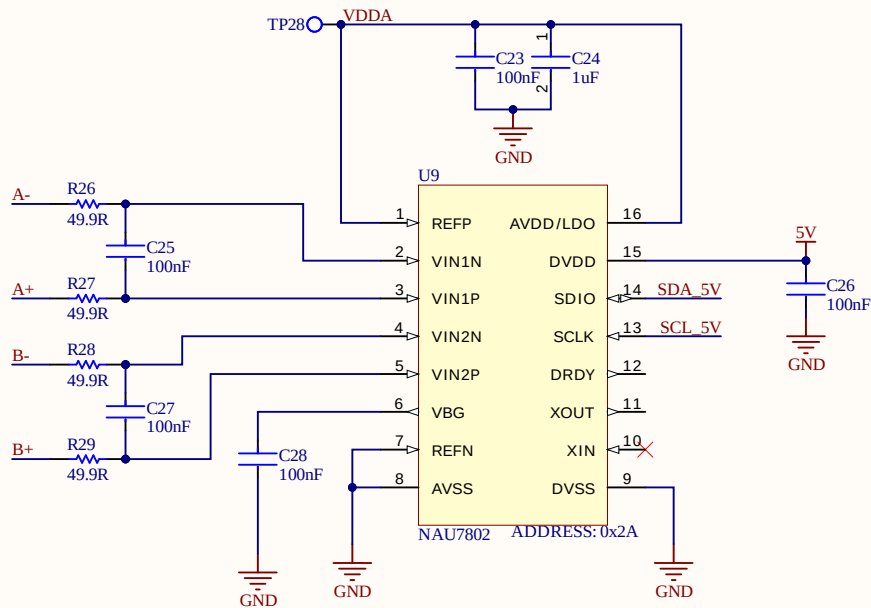
Harness Entry



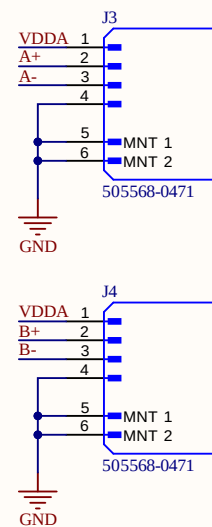
Level Shifting



Loadcell Frontend



External Interface



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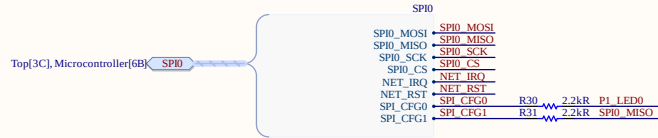
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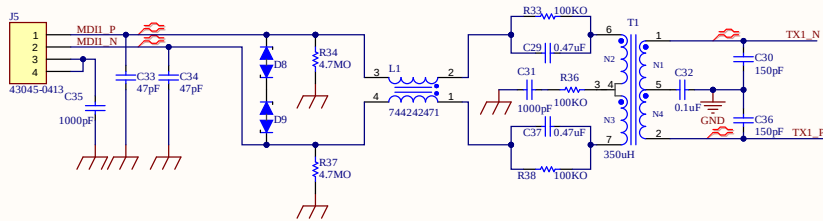
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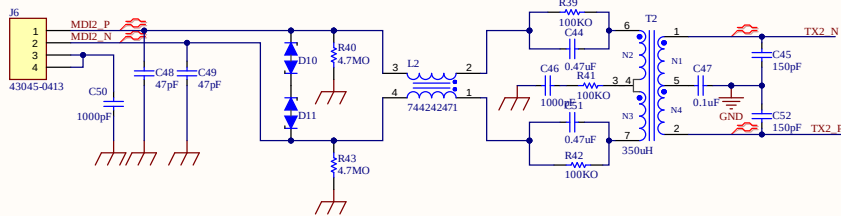
Harness Entry



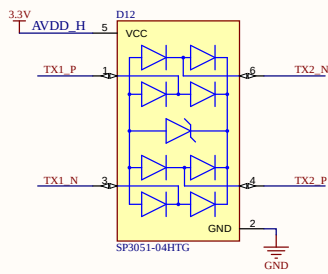
MDI Channel 1



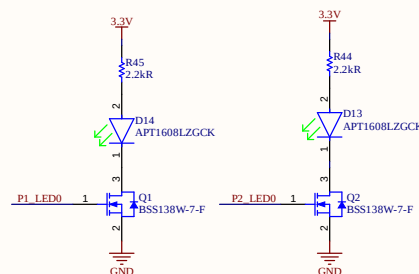
MDI Channel 2



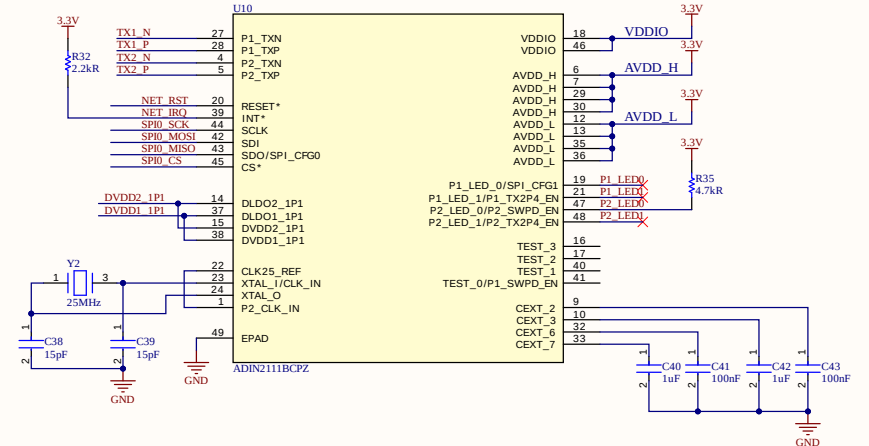
ESD Protection



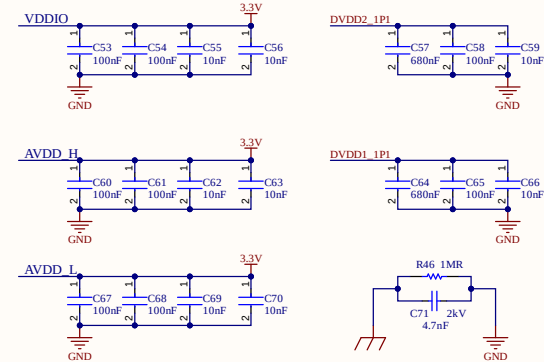
Indicator LEDs



ADIN2111



Decoupling



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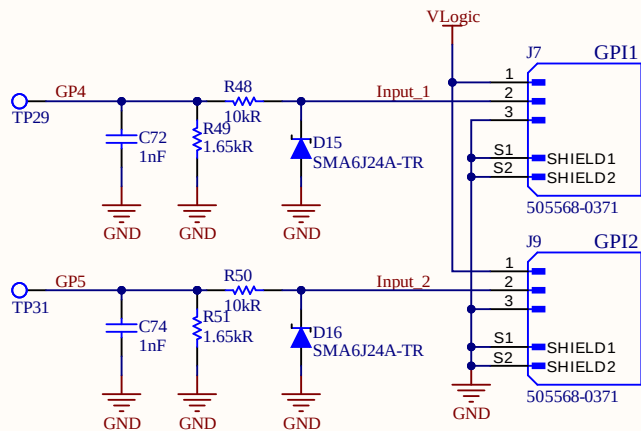
ETHERNET

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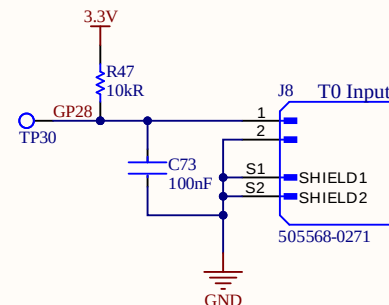
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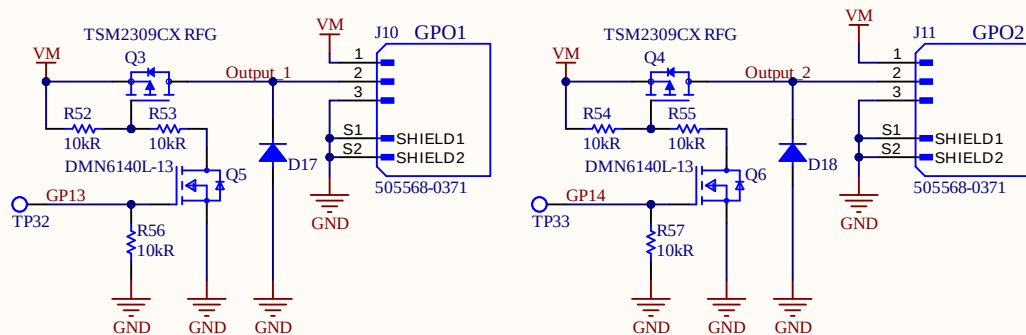
24V Inputs



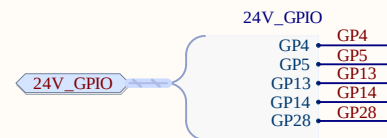
Thermistor Input



24V Outputs



Harness entry



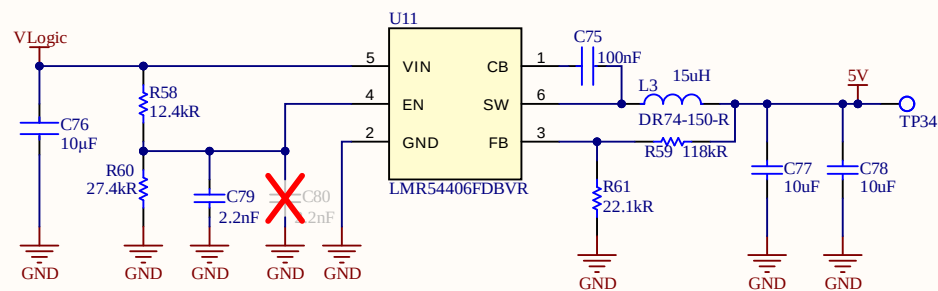
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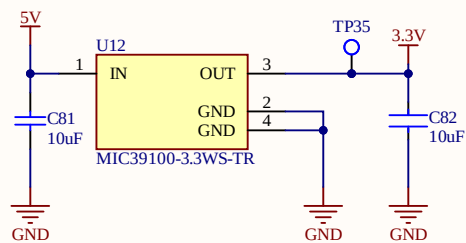
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5V Switching Regulator



3.3V Linear Regulator



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